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MILITARY SPECIFICATION

CARTRIDGE, BOMB EJECTION MLU-11/C
AND
CARTRIDGE KIT, BOMB EJECTION MLU-12/C

1. SCOPE

1.1 This specification covers a component designated as Cartridge, Bomb Ejection MLU-11/C hereafter referred to as the cartridge, and a combination of two cartridges and container designated as the Cartridge Kit, Bomb Ejection MLU-12/C hereafter referred to as the kit.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-5424	Cable: Steel, (Corrosion-Resisting), Flexible, Preformed (For Aeronautical Use)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order Of Precedence for the Selection Of
MIL-STD-709	Ammunition Color Coding
MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Water Proofing, with Appropriate Test Methods.

FSC 1377

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DRAWINGS

Air Force

61D14605	Cable Cutter MHU-29/C Clip-IN Subassembly
61D14650	Cartridge Kit, Bomb Ejection MLU-12/C
62D13001	Cartridge, Bomb Ejection MLU-11/C

PUBLICATIONS

Department of Defense

Index of Specifications and Standards

Air Force Specification Bulletin

23 Material and Process Specifications

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49CFR71-78 Interstate Commerce Commission Regulations for the Transportation of Explosives and Other Dangerous Material

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulation (1949 Editions - Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite 49CFR71-78 (Revised 1950).

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of Specifications and Standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143; provided that, for materials and processes, the applicable Group I items per MIL-STD-143, shall be selected first from USAF Specification Bulletin 23, and if not included herein, then from the Department of Defense Index of Specifications and Standards.

3.3 Material. Material shall be selected and processed for the various parts in accordance with applicable specifications and standards as indicated on the drawings.

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3.4 Design and construction.

3.4.1 Cartridge. The MLU-11/C Cartridge shall be designed and constructed in conformance to Air Force drawing 62D13001.

3.4.2 Cartridge kit. The MLU-12/C Cartridge Kit shall be designed and constructed in conformance to drawing 61D14650.

3.5 Function. When fired in a 61D14605 cable cutter, a single MLU-11/C Cartridge shall generate sufficient force to actuate the cutter in a manner such as to sever a 3/8 x 7 x 19 corrosion resistant cable which meets the requirements of MIL-C-5424. This requirement shall be met with the cable under any tensile load from zero to 5500 pounds.

3.5.1 All-fire sensitivity. The cartridge shall be capable of being fired with a minimum direct current of 5 amperes under all environmental conditions.

3.5.2 No-fire sensitivity. The cartridge shall be capable of withstanding a maximum direct current of 2 amperes for a duration of 5 minutes without firing.

3.5.3 Firing time. When subjected to a firing current of 5 plus 0.2, minus 0.0 amperes, the cartridge shall fire within 5 to 20 milliseconds under standard conditions, and from 5 to 40 milliseconds under all environmental conditions.

3.5.4 Electrostatic sensitivity. The cartridge shall be capable of firing after repeated discharges of 200,000 ergs from a 500 picofarad capacitor.

3.5.5 Non-shorting. The bridgewire shall not become shorted as a result of firing the cartridge.

3.6 Environmental requirements.

3.6.1 Acoustic power. The cartridge shall not fire when subjected to acoustical power levels up to 140 decibels for 50 test cycles, each cycle of one minute duration.

3.6.2 Acceleration. The cartridge shall be capable of proper operation after sustaining a 50g acceleration for at least one minute in each of its three mutually perpendicular axes.

3.6.3 Impact shock. The cartridge shall be capable of proper operation after being subjected to an 8-foot drop. The cartridge shall not fire due to a drop of 40 feet.

3.6.4 High temperature. The cartridge shall be capable of proper operation after prolonged exposure to a temperature of 121°C (250°F). The cartridge shall not fire during prolonged exposure to a temperature of 143°C (290°F).

3.6.5 Low temperature. The cartridge shall be capable of proper operation after prolonged exposure at -67°C (-89°F).

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3.6.6 Temperature shock. The cartridge shall be capable of proper operation after being subjected to the temperature shock testing requirements of this specification.

3.6.7 Vibration. The cartridge shall be capable of satisfactory operation during and after vibration, as specified in vibration tests of this specification.

3.6.8 Fluid leakage. The cartridge shall be capable of satisfactory operation after being subjected to immersion in water or aircraft hydraulic fluid.

3.6.9 Altitude. The cartridge shall be capable of proper operation during and after exposure to a temperature of -54°C (-65°F) and to a pressure of 93.672 pounds per square foot (70,000 feet above sea level).

3.6.10 Humidity. The cartridge shall be capable of satisfactory operation after subjection to humidity likely to be encountered in tropical climates.

3.6.11 Salt spray. The cartridge shall be capable of proper operation after prolonged exposure to a salt sea atmosphere and spray.

3.6.12 Fungus. The cartridge shall be capable of proper operation after being subjected to fungus growth as encountered in tropical climates.

3.6.13 Sand, dust and ice. The cartridge shall be capable of proper operation after exposure to sand, dust and ice.

3.6.14 Explosion Proof. When the cartridge is fired in a 61D14605 cable cutter under the conditions specified in 3.5 and surrounded by an explosive atmosphere, initiation of an explosion shall not occur.

3.7 Identification of product. The cartridge and kit shall be marked for identification in accordance with MIL-STD-130, and shall show the additional information specified in 3.7.1 and 3.7.2.

3.7.1 Lot numbering. Each separate production lot of the cartridge shall be assigned an ammunition lot number in accordance with MIL-STD-1168. This lot number shall be marked on each cartridge.

3.7.2 Color coding. The color marking of the cartridge and kit shall conform to the engineering drawings and MIL-STD-709.

3.8 ICC explosive classification. MLU-11/C Cartridges and MLU-12/C Kits shall be assigned the following ICC explosive classifications:

a. 1000 or more MLU-11/C Cartridges or 500 or more MLU-12/C Kits - Class "A".

b. Less than 1000 MLU-11/C Cartridges or less than 500 MLU-12/C Kits - Class "C".

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3.8.1 Quantity distance classification. The cartridges or kits shall be in Class 8, Compatibility Group P for storage and quantity distance purposes.

3.9 Ammunition data card. Ammunition Data Cards shall be initiated and maintained for each lot of kits in accordance with MIL-STD-1167. Specific information pertinent to common items such as screws and washers may be omitted.

3.10 Government furnished property. When specified in the contract, the Government will furnish the following to the contractor, upon his request, for use in making the MLU-11/C Cartridge.

Item No.	Part No.	Description of Item	Quantity
1	61D14605	Cable Cutter, MHU-29/C Clip-In Subassembly	1 per each 10 firings or portion thereof, which require installation of cartridge in cable cutter

3.11 Workmanship. The cartridge, including all parts and accessories, shall be fabricated in a manner that will insure that the criteria of appearance, fit, and adherence to specified tolerances shall be observed. Particular attention shall be given to the markings required under "Identification of Product".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction testing.

4.2.1 Preproduction test samples. The preproduction test samples shall be representative of production cartridges and shall have passed the individual tests. They shall be consecutively numbered for test identification. They shall be tested at a laboratory designated by the procuring activity, or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.1.1 Preproduction test report. In the event that preproduction testing is performed at the contractor's plant, the contractor shall, after completion of the preproduction tests, prepare a preproduction test report according to MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

4.2.2 Preproduction tests. Preproduction tests shall consist of all tests described herein under Preproduction Test Methods.

4.2.3 Preproduction schedule and test plan. The preproduction schedule and test plan shall be as described in Preproduction Test Methods.

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4.3 Acceptance tests. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Plan and Tests

4.3.1 Individual tests. Each cartridge shall be subjected to the following tests as described under Individual Test Methods of this specification:

a. Examination of Product

b. Conformance to Drawings

c. Electrical Resistance Tests

4.3.2 Sampling plan and tests.

4.3.2.1 Lot. Lot definition, size, and formation shall be in accordance with MIL-STD-105.

4.3.2.2 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105. Inspection level I shall be used. For production lots of less than 800 the AQL shall be 1.5. For production lots of 800 or more, the AQL shall be 1.0.

4.3.2.2.1 Sampling tests. The samples selected shall be subjected to the sampling procedures of the following tests as described in Sampling Test Methods of this Specification:

a. Temperature Shock Test

b. Vibration Test

c. Linear Acceleration Test

d. Fluid Leakage Test

4.3.2.3 Individual tests may continue. For operational reasons, individual tests may be continued pending investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that items meet all the requirements of this Specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test conditions.

4.4.1 Standard conditions. Standard conditions for conducting the tests shall be as follows:

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a. Temperature 21°C (70°F) plus or minus 11°C (20°F).

b. Relative Humidity 90 percent or less.

4.4.2 Test temperature. The cartridge shall be stablized at the specified temperature prior to conducting any tests. Temperature stabilization shall be as defined in MIL-E-5272.

4.4.3 Test measurements.

4.4.3.1 Instruments. All measurements shall be made with instruments whose accuracy has been verified. Calibration records shall be maintained.

4.4.3.2 Tolerances. The maximum allowable tolerances on test condition measurements shall be as follows:

a. Temperature - plus or minus 5°F.

b. Relative Humidity - plus or minus 5 percent.

c. Vibration and Shock Amplitude - plus or minus 5 percent.

d. Vibration Frequency - plus or minus 2 percent.

e. Current - plus or minus 2 percent.

f. Resistance - plus or minus 2 percent.

g. Time Interval - plus or minus 2 percent.

h. Sound Level - plus or minus 1 decibel.

4.4.3.3 Required measurements (for each firing).

a. Resistance or unit prior to firing.

b. Firing current.

c. Firing time.

d. For preproduction testing, the cartridge number shall be recorded.

4.5 Suggested test equipment.

a. Wheatstone Bridge capable of measuring 0 to 1.0 ohms with an accuracy of plus or minus 0.001 ohm and an output to the cartridge of not more than 0.005 ampere.

b. Ammeter, 0 to 10 amperes direct current, with an accuracy of plus or minus 0.5 percent.

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c. Oscillograph with 850 cycle or higher galvanometers, and capable of a paper speed of at least 90 inches per second.

d. Carbon Microphone Button.

e. Electrostatic Voltmeter, 0 to 10 kilovolt range.

f. Van de Graaff Generator capable of developing 15 kilovolts.

g. Cartridge holding fixture with provisions for containment of explosion and fragmentation.

4.6 Test methods.

4.6.1 Examination of product. Each cartridge shall be given a complete visual examination to determine compliance with the requirements of this specification, including workmanship.

4.6.2 Conformance to drawings. Each cartridge shall be inspected for strict conformance to the drawings and specifications and standards referenced therein.

4.6.3 Electrical resistance tests. Each cartridge shall be tested with a Wheatstone Bridge with current output limited to not more than 0.005 ampere to show conformance to the electrical resistance requirements on drawing 62D13001.

4.6.4 Preproduction test methods.

4.6.4.1 Preproduction schedule and test plan. The preproduction schedule and test plan shall be as shown in Tables I and II. A total of 204 cartridges shall be required for preproduction testing, and shall be divided into ten lots in accordance with Tables I and II. Each lot shall contain consecutively fabricated cartridges, and shall be divided by their order of fabrication. Twenty cartridges shall be programmed as spares. (NOTE: It is recommended that a sufficient quantity of cartridges be manufactured, in the same lot, to compensate for units which may be rejected prior to testing.)

4.6.4.2 Preproduction separate tests. Whenever any of the separate tests of Table I require firing the cartridge, each cartridge subjected to the test shall be fired. The electrical circuit for the firing tests shall be in accordance with figure 1.

4.6.4.3 Preproduction consecutive tests. Upon completion of each of the consecutive tests of Table II, resistance of each cartridge shall be recorded and ten cartridges shall be withdrawn, one from each lot. Six of the withdrawn cartridges shall be fired, and four of the cartridges shall be subjected to the No-Fire Sensitivity Test. The electrical circuit for the firing tests shall be in accordance with figure 1. Upon completion of the Fluid Leakage Test, ten of the remaining 20 cartridges shall be fired. The direct current shall be 5.0 plus 0.20 minus 0.0 amperes. The last ten cartridges shall be subjected to the No-Fire Sensitivity Test. The electrical circuit for the firing tests shall be in accordance with figure 1.

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4.6.4.4 Failure during separate testing. In the event of failure of one of the cartridges to meet electrical and performance requirements while being subjected to the Separate Tests, ten cartridges from the 20 spares shall be selected, one from each lot, and subjected to the test in which failure occurred. If the ten cartridges function as required, the cartridges shall be considered to have met the preproduction requirements. No more than one failure will be acceptable.

4.6.4.5 Firing during consecutive testing. In the event of failure of one of the cartridges to meet electrical and performance requirements while being subjected to the Consecutive Tests, testing shall continue as specified. Ten spares shall be subjected to the test in which failure occurred, and to all preceding tests in the order specified by the Test Schedule. Upon successful completion by the ten spares of the test in which a failure occurred, the cartridges shall be considered to have passed the consecutive tests of the preproduction test program. No more than one failure shall be acceptable.

4.6.4.6 All-fire sensitivity. Ten cartridges shall be tested at standard conditions. The cartridges shall be fired in a test jig. The direct current shall be 5 amperes. No failure shall occur. The firing time shall not exceed the specified time limits.

4.6.4.7 No-fire sensitivity. Five cartridges shall be tested at standard conditions. The cartridges shall be held in a test jig. The direct current shall be 2 amperes for a duration of five minutes. The cartridges shall not fire. They are not required to fire after this period. (NOTE: The number of cartridges to be tested specified above, applies to the Separate Preproduction Tests only. When this procedure is followed for Preproduction Consecutive Tests, see Table II for quantities.)

4.6.4.8 Electrostatic discharge operation. A 200,000 erg discharge from a 500 picofarad condenser shall be applied to the contacts of five cartridges. The electrical circuit shall be in accordance with figure 2. Ten pulses shall be applied to each cartridge, allowing sufficient time between pulses for the cartridge circuit to cool. The cartridges shall not fire. Each unit shall then be fired.

4.6.4.9 Acoustical power. Five cartridges shall be subjected to the acoustical power levels of figure 3 for 50 one-minute cycles. The cartridges shall not fire during the test. At the end of the test the cartridges shall be fired.

4.6.4.10 Functioning test. Functional testing of the cartridge shall be performed in the following manner:

a. Assemble a length of 3/8 x 7 x 19 cable conforming to MIL-C-5424 in a 61D14605 Cable Cutter. There shall be no tensile load on the cable. The cable shall be of sufficient length to extend beyond both sides of the cable cutter housing.

b. Install a single cartridge in the cable cutter.

c. Unlock the cable cutter and fire the cartridge.

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SEPARATE TESTS	TOTAL	PREPRODUCTION TEST LOTS 1	2	3	4	5	6	7	8	9	10
All Fire	10	1	1	1	1	1	1	1	1	1	1
No Fire	5	1		1		1		1		1	
Electro Static Discharge	5		1		1		1		1		1
Acoustical Power	5	1		1		1		1	1		
Functioning	3			1		1		1			
Linear Acceleration	5	1		1		1		1		1	
Impact Shock Operation	6	1	1	1		1		1		1	
Impact Shock Hazard	6		1		1		1		1	1	1
High Temperature Operation	5		1		1		1		1		1
Low Temperature	5		1		1		1		1		1
High Temperature Hazard	5	1		1		1		1		1	
Temperature Shock	10	1	1	1	1	1	1	1	1	1	1
Vibration a.	5		1		1		1		1		1
Vibration b.	5	1		1		1		1		1	
Fluid Leakage (Water)	5		1		1		1		1		1
Fluid Leakage (Hydraulic)	5	1		1		1		1		1	
Altitude	5	1		1		1		1		1	
Fungus	5	1		1		1		1		1	
Humidity	5		1		1		1		1		1
Sand and Dust	5		1		1		1		1		1
Ice	5	1		1		1		1		1	
Non-Shorting	*5	1			1		1		1		1
Salt Spray	5		1		1		1		1		1
Explosion Proof	4		1		1		1				1
TOTAL	124	12	13	13	12	13	12	13	12	12	12

*These cartridges fired in previous firing tests.

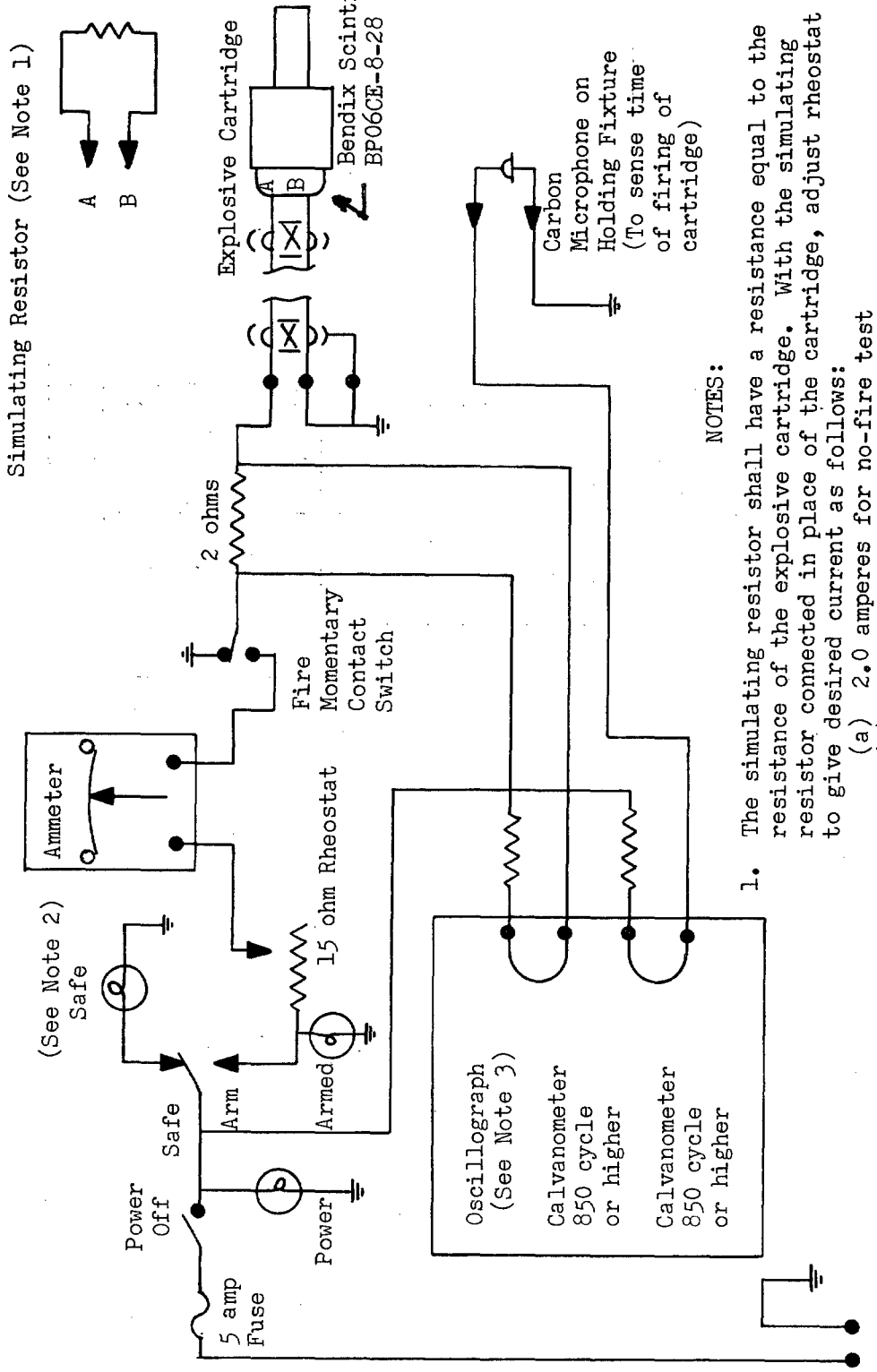
TABLE 1 - PREPRODUCTION SEPARATE TEST SCHEDULE

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CONSECUTIVE TESTS	TOTAL TEST	1	2	3	4	5	6	7	8	9	10
Temperature Shock	60	6	6	6	6	6	6	6	6	6	6
Vibration b.	50	5	5	5	5	5	5	5	5	5	5
Linear Acceleration	40	4	4	4	4	4	4	4	4	4	4
Impact Shock Op	30	3	3	3	3	3	3	3	3	3	3
Fluid Leakage (H_2O)	20	2	2	2	2	2	2	2	2	2	2
No Fire Sensitivity	10	1	1	1	1	1	1	1	1	1	1

TABLE II - PREPRODUCTION CONSECUTIVE TEST SCHEDULE

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NOTES:

1. The simulating resistor shall have a resistance equal to the resistance of the explosive cartridge. With the simulating resistor connected in place of the cartridge, adjust rheostat to give desired current as follows:
   (a) 2.0 amperes for no-fire test
   (b) 5.0 amperes for all fire test
   (c) 5.0 plus 0.2 minus 0.0 amperes for all other tests.
2. When connecting live cartridge, caution should be exercised to assure that firing circuit is in "safe" position.
3. Oscillograph shall record time of signal input and time of separation of bridge wire of cartridge. The 850 cycle galvanometer will provide the required response time for the recording equipment.

FIGURE 1. FIRING AND RECORDING CIRCUITRY

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d. Inspect and determine that the cable has been completely severed and that no physical damage to the cable cutter has occurred.

e. Repeat steps a, b, and c, except that two cartridges shall be installed in the cable cutter and fired.

f. Repeat step d.

4.6.4.11 Linear acceleration. Five cartridges shall be secured to a centrifuge successively in six positions such that each direction along each of the three mutually perpendicular major axes in turn extends in a radial direction with respect to the centrifuge center of rotation, and shall be subjected to a sustained acceleration of 50g for at least one minute in each direction. After completion of the above, the cartridges shall be fired. (NOTE: The number o f cartridges to be tested specified above, applies to the Separate Preproduction Tests only. When this procedure is followed for Preproduction Consecutive Tests, only. See Table II for quantities.)

4.6.4.12 Impact shock operation. Six cartridges shall be dropped from a height of eight feet onto a steel plate supported by concrete. The attitude at impact and number of drops shall be as follows:

	Attitude	Number of Cartridges
a.	Cartridge nose down	2
b.	Cartridge nose up	2
c.	Cartridge horizontal	2

The cartridges shall not fire. Each cartridge shall then be fired.

4.6.4.13 Impact shock handling. Two each cartridges shall be dropped from a height of 40 feet in the nose-down vertical position; two each cartridges in the nose-up vertical position; and two each cartridges in the horizontal position. Each cartridge shall be dropped onto a steel plate solidly supported by a rigid concrete base. The cartridges shall not fire. It is not required that the units be usable after the drop test.

4.6.4.14 High temperature operation. Five cartridges shall be conditioned at a temperature of 121°C (250°F) for 50 hours. After the 50-hour period and while at this temperature, the cartridges shall be fired.

4.6.4.15 High temperature hazard. Five cartridges shall be conditioned at a temperature of 143°C (290°F) for a period of 360 hours. The cartridges shall not fire during the test. They are not required to fire after this period.

4.6.4.16 Low temperature storage. Five cartridges shall be conditioned at a temperature of -67°C (-89°F) for a period of 24 hours. At the end of this time and while at this temperature, the cartridges shall be fired.

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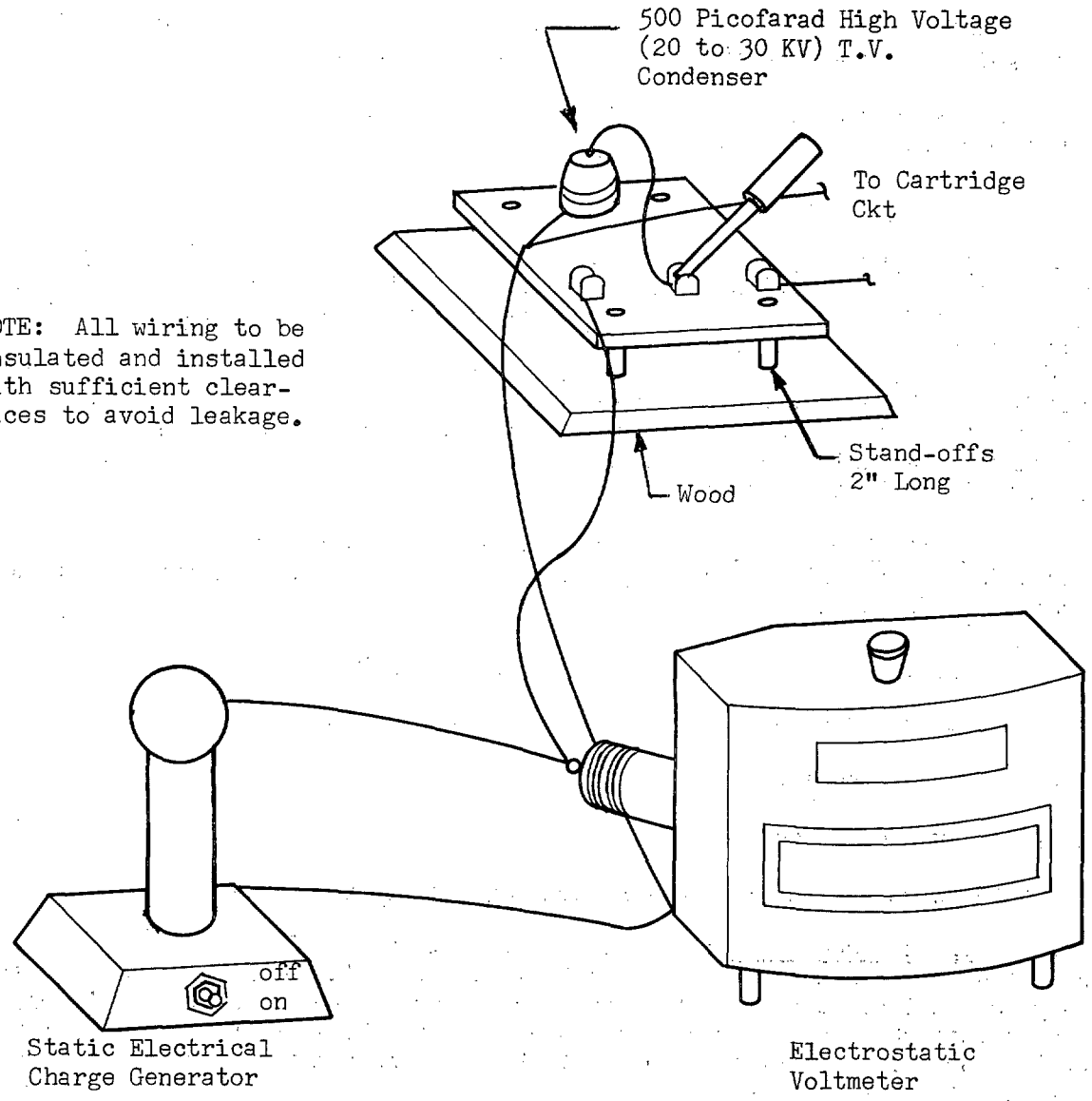


FIGURE 2. ELECTROSTATIC DISCHARGE TEST SET-UP

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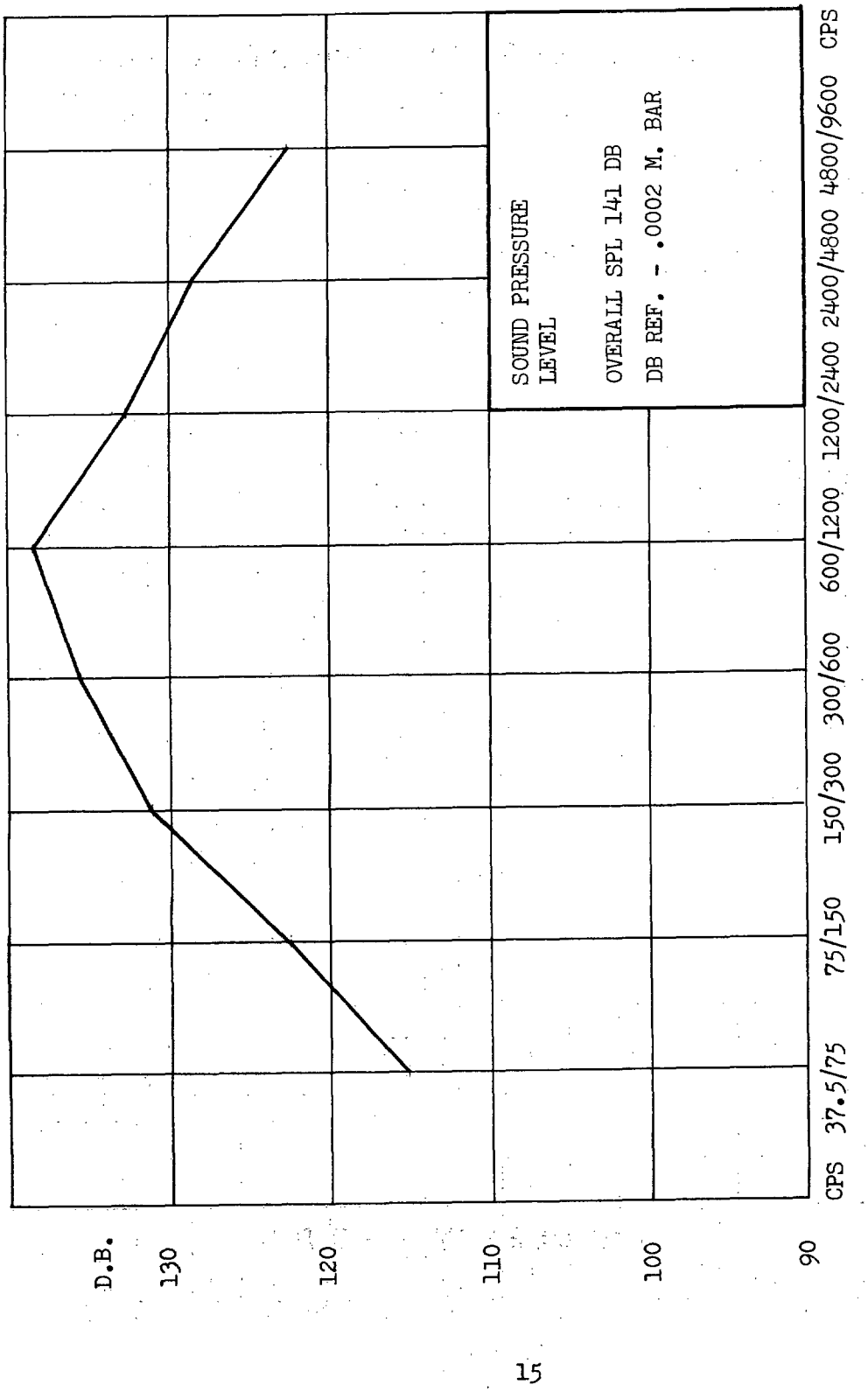


FIGURE 3

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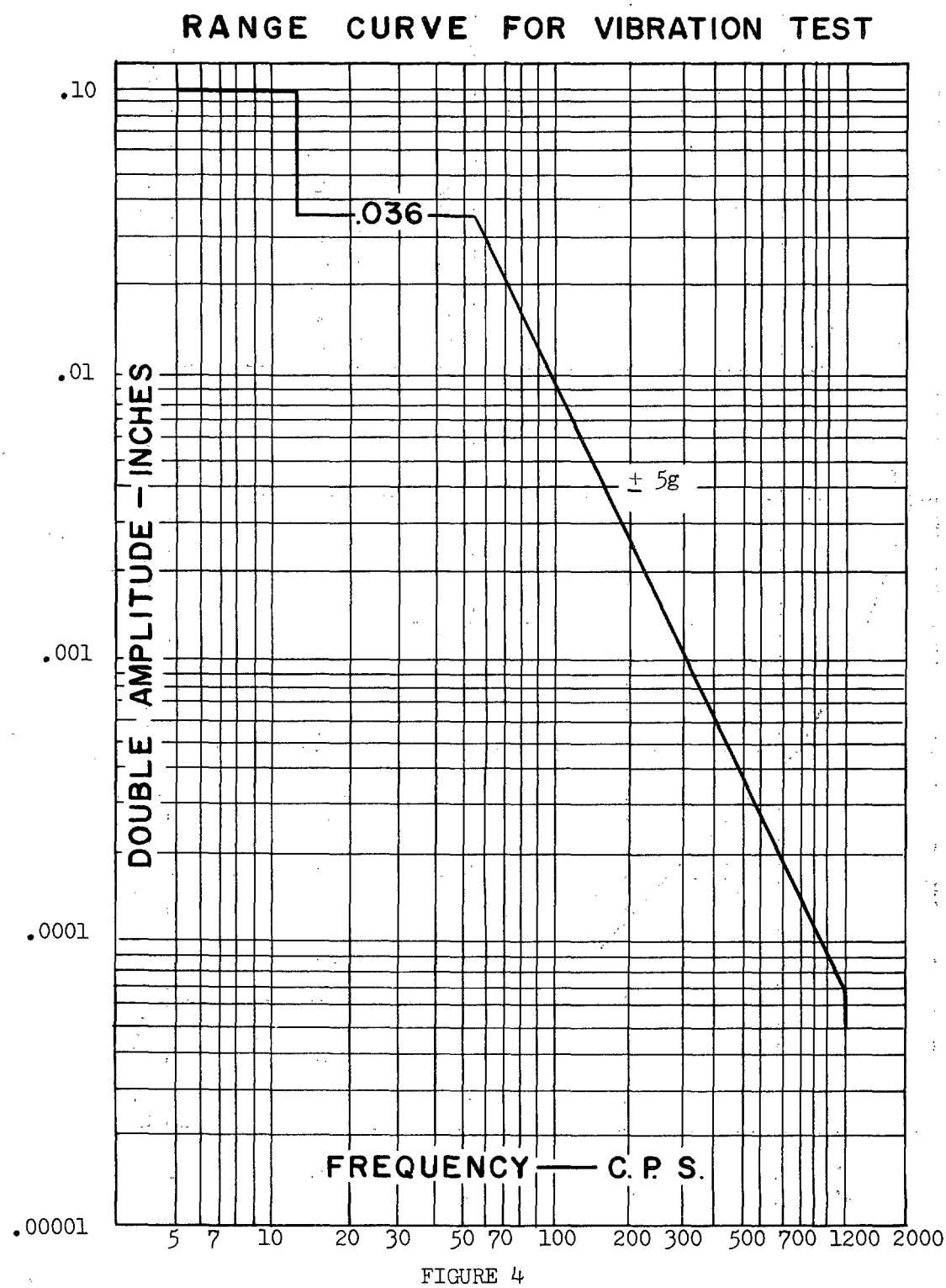


FIGURE 4

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4.6.4.17 Temperature shock. Ten cartridges shall be fired following 16 cycles of temperature shock. The cartridges shall be placed in a chamber stabilized at -54°C (-65°F) for four hours and then transferred to a chamber stabilized at 121°C (250°F) for four hours for each cycle. The transfer time from chamber to chamber shall not exceed five minutes. (NOTE: The number of cartridges to be tested specified above, applies to the Separate Preproduction Tests only. When this procedure is followed for Preproduction Consecutive Tests, see Table II for quantities.) When this procedure is followed for Preproduction Consecutive Tests the cartridges shall be subjected to eight cycles of temperature shock.

4.6.4.18 Vibration

4.6.4.18.1 Separate preproduction vibration tests. Ten cartridges shall be subjected to vibration tests as follows and in the order listed:

Cycling. The cartridges shall be vibrated under the cycling conditions specified herein for the periods listed in the Vibration Test Schedule. The applied cycling shall be between 50 and 1200 cycles per second in 15 minute cycles at vibratory accelerations as shown in figure 4. The vibration shall be applied along each of the cartridge's three mutually perpendicular major axes according to the Vibration Test Schedule.

Vibration Test Schedule

(Times shown refer to one axis of vibration)

Type	Vibration (Ambient Conditions)	Vibration 71°C (160°F)	Vibration -54°C (-65°F)
Cycling	60 minutes	15 minutes	15 minutes
Resonance	60 minutes	15 minutes	15 minutes

Resonance. Resonant frequencies of the cartridge shall be determined by varying the frequency of applied vibration slowly through the specified range of frequencies at vibratory accelerations not exceeding those in figure 4. Individual resonant frequency surveys shall be conducted with the vibration applied along each of the three mutually perpendicular axes of the cartridge. The cartridge shall be vibrated at the indicated resonant conditions for the periods shown in the vibration test schedule and with the applied double amplitude or vibratory acceleration specified in figure 4. These periods of vibration are applied along each of the three mutually perpendicular major axes of vibration. When more than one resonant frequency is encountered with vibration applied along any one axis, the test period shall be accomplished at the most severe resonance. If no resonant frequency is found, a frequency of 100 cycles per second shall be used. The cartridges shall be fired as follows:

a. Five cartridges shall be fired at the completion of the resonant frequency vibration test while vibrating at resonant frequency.

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b. Five cartridges shall be fired subsequent to the completion of the vibration tests.

4.6.4.18.2 Consecutive preproduction vibration tests. See Table II for quantities. The cartridges shall be subjected to the resonance test of 4.6.4.18.1 only.

4.6.4.19 Fluid leakage resistance.

4.6.4.19.1 Water. Five cartridges shall be immersed in water which has a temperature of 10°C (50°F) and to a depth no greater than three inches after immersion. The absolute pressure of the air above the liquid shall then be reduced to 0.25 pounds per square inch absolute until air bubbles substantially cease to be given off by the water. The absolute pressure shall then be increased to 0.50 pounds per square inch. Any bubbles coming from within the cartridge case shall be considered as leakage. Bubbles which are the result of entrapped air on the various exterior parts of the case shall not be considered as leakage. The pressure within the test chamber shall then be increased to atmospheric pressure and the cartridges shall remain immersed for a period of 15 minutes. At the conclusion of that period of time, the cartridges shall be removed and fired.

4.6.4.19.2 Hydraulic fluid. Five cartridges shall be subjected to the conditions stated in 4.6.4.19.1 except that the liquid shall be aircraft hydraulic fluid.

NOTE #1: Although the appearance of bubbles shall be regarded as leakage, firing tests results comparable to those obtained with cartridges fired at standard conditions shall indicate that such leakage is not cause for rejection.

NOTE #2: The number of cartridges to be tested specified above applies to the Separate Preproduction Tests only. When this procedure is followed for Preproduction Consecutive Tests, see Table II for quantities.

4.6.4.20 Altitude operation. Five cartridges shall be conditioned at -54°C (-65°F). The chamber pressure shall then be reduced to 93.672 lbs per sq foot (70,000) feet above sea level). These conditions shall be maintained for four hours. At the end of this time the cartridges shall be fired.

4.6.4.21 Humidity and condensation. Five cartridges shall be placed in a test chamber having a temperature and relative humidity of 120°F and 90 percent respectively. These conditions shall be maintained for 360 hours. At the conclusion of this period, the cartridges shall be removed and fired.

4.6.4.22 Salt spray. Five cartridges shall be tested according to procedure I of MIL-E-5272 except that the length of the salt spray test shall be 50 hours. At the end of the test period, the cartridges shall be fired.

4.6.4.23 Fungus. Five cartridges shall be subjected to tests in accordance with procedure I of MIL-E-5272. Upon completion of the tests, the cartridges shall be examined visually for evidence of deterioration and fired. The

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cartridges subjected to this test shall not have previously been subjected to the Salt Spray or Sand and Dust tests.

4.6.4.24 Sand and dust. Five cartridges, with their connectors protected by dust proof caps, shall be tested in accordance with procedure I of MIL-E-5272, except that the sand velocity shall be 2500 ± 500 feet per minute. Upon completion of Part II, the cartridges shall be removed, visually inspected for dust penetration, and fired.

4.6.4.25 Ice. Five cartridges shall be placed in a test chamber at -57°C (-70°F) and stabilized. The cartridges shall then be removed and exposed to an atmosphere having a temperature of approximately 38°C (100°F), and a relative humidity of approximately 90%, until all evidence of frost has disappeared. Retaining all the condensation practicable, the units shall be replaced in the chamber at -57°C (-70°F). There shall be three cycles from 38°C to -57°C. At the end of the last cycle while at -57°C, the units shall be removed and fired.

4.6.4.26 Non-shorting. Upon completion of the separate and consecutive tests, ten fired units shall be selected, placed in a test chamber, and the pressure reduced to .16 pounds per square inch. A voltage, sufficient to produce a 200 ampere current in the bridgewire of an unfired cartridge, shall be applied to the terminal pins of each cartridge. The cartridges shall be non-shorting.

4.6.4.27 Explosion proof. The cartridges shall be subjected to an explosion proof test in accordance with procedure III, MIL-E-5272 with the following exceptions:

a. The fuel used shall be equivalent to, or more flammable than the optimum mixture of JP-4 fuel and air. Commercial butane or propane are satisfactory fuels for this test.

b. This test shall be performed at one simulated altitude only. This altitude shall be ground level of the testing laboratory.

4.6.4.27.1 Firing procedure. Firing of the sample cartridges during the explosion proof test shall be accomplished as follows:

a. Repeat steps a., b., and c. of the Functioning Test. (4.6.4.10)

b. Repeat steps a., b., and c. of the Functioning Test, (4.6.4.10) except that the cartridges shall be installed and fired in a chamber not used in a. above.

c. Repeat steps a., b., and c. of the Functioning Test, (4.6.4.10) except that two cartridges shall be installed in the cable cutter and fired in the explosive mixture.

d. Repeat step d. of the Functioning Test. (4.6.4.10)

4.6.5 Sampling test methods. Samples shall be subjected to all sampling tests in the order listed below and at the end of each test the resistance of each

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cartridge shall be recorded. Upon completion of these tests, the samples shall be separated into two groups. Group I shall be fired. The electrical test circuit shall be in accordance with figure 1. Group II shall be subjected to the No Fire Sensitivity Test. Each group shall contain a minimum of five cartridges.

4.6.5.1 Temperature shock. Cartridges shall be subjected to one cycle of temperature shock. Cartridges shall be placed in a chamber stabilized at a temperature of -54°C (-65°F) for four hours and then transferred to a chamber stabilized at 121°C (250°F) for four hours to complete the cycle. The transfer time from chamber to chamber shall not exceed five minutes. The cartridge shall be removed from the hot chamber and allowed to cool to standard temperature.

4.6.5.2 Vibration. Cartridges shall be vibrated at the most severe resonant frequency, as determined by the preproduction vibration test, for a period of 15 minutes. If no resonant frequencies are apparent within the specified frequency range, the cartridges shall be vibrated at a frequency of 100 cycles per second and an applied double amplitude of 0.009 inch for a period of 15 minutes.

4.6.5.3 Linear acceleration. Cartridges shall be secured to a centrifuge such that the longitudinal axis and the explosive end of the cartridge extend in a radial direction with respect to the centrifuge center of rotation. The cartridges shall be subjected to a sustained acceleration of 50g for at least one minute in this position.

4.6.5.4 Fluid leakage. Cartridges shall be immersed in water which has a temperature of 10°C (50°F) and to a depth no greater than three inches after immersion. The absolute pressure of the air above the liquid shall be reduced to 0.25 pounds per square inch absolute until air bubbles substantially cease to be given off by the water. The absolute pressure shall then be increased to 0.50 pounds per square inch. Any bubbles coming from within the cartridge case shall be considered as leakage. Bubbles which are the result of entrapped air on the various exterior parts of the case shall not be considered as leakage. The pressure within the test chamber shall then be increased to atmospheric pressure and the cartridges shall remain immersed for a period of 15 minutes.

4.7 Basis for rejection.

a. Individual units failing to meet their respective tests shall be rejected. Rejected units either shall be replaced or shall be reworked to correct the defects, after which all specified tests shall be repeated. Rejected units shall not be resubmitted for inspection without furnishing all particulars concerning previous rejection and measures taken to overcome the defects.

b. All parts, specimens or assemblies destroyed in making tests required by this specification shall be in addition to the quantity specified in the order and shall be furnished to the procuring activity without increasing the cost of the contract. Accepted units shall have such acceptance marking on the units as specified by the procuring activity.

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c. After any test, the occurrence of any of the following conditions shall be sufficient cause for rejection:

(1) Failure to perform within required tolerances.

(2) Loose connections.

(3) Badly soldered joints.

(4) Evidence of sufficient corrosion to interfere with fit, insertion in or removal from Clip-In Subassembly.

5. PREPARATION FOR DELIVERY

5.1 Packaging. For all levels of shipment, two cartridges shall be packaged in accordance with drawing 61D14650. When so packaged, the cartridges and sealed container shall constitute the Cartridge Kit, Bomb Ejection MLU-12/C.

5.2 Packing. For all levels of shipment, a number of kits specified by the procuring activity shall be packed in a box conforming to PPP-B-621. Physical protection shall conform to MIL-STD-1186.

5.3 Marking. Exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

CARTRIDGE KIT, BOMB EJECTION MLU-12/C

5.3.1 Additional marking (explosive). In addition to marking in accordance with MIL-STD-129, the exterior containers shall be marked in accordance with Code of Federal Regulations 49CFR71-78. The following specific warning shall appear on the exterior container: "ELECTRIC EXPLOSIVE CARTRIDGES, Handle with Care."

6. NOTES

6.1 Intended use. The Cartridge, Bomb Ejection MLU-11/C is intended for use in the release mechanism of clip-in subassemblies for single weapon suspension, and the Cartridge Kit, Bomb Ejection MLU-12/C is intended to provide an adequately protected means of including the proper number of cartridges for each release mechanism.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Whether preproduction tests will be conducted by procurement activity.

6.2.1 Preproduction samples. It is expected that the contract or purchase order will specify that 204 cartridges, representative of the production cartridges will be submitted to the procuring activity for test and evaluation. The invitation for bids, and the contract or purchase order should specify the point of delivery for these items.

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6.3 Production delivery. No production items should be delivered prior to approval of the preproduction samples.

6.4 Definitions.

a. Preproduction testing. Preproduction tests are those tests made on samples which have been submitted by a contractor on a production contract or purchase order, to prove, prior to his starting production, that the equipment meets the requirements of this specification.

b. Acceptance tests. Acceptance tests are those tests accomplished on equipment submitted for acceptance under a contract to determine that the units are equivalent in performance and construction to the approved preproduction samples.

6.5 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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